LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.
5. Specific Objectives	17.1.5.1 Trainee will be able to explain the construction and functioning of aircraft flight controls systems and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Classroom discussion on control system types; Safety: Maintain discipline; Supervision: Instructor leads.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft flight controls?
(b) Name the three main categories of aircraft control systems.
(c) How does a mechanical control system transmit pilot input?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Mechanical Control System Construction • Power Assisted Control System Operation • Fully Powered Control System Features • Fly-by-Wire System Architecture • Comparison of Control System Types	L & Ds	PPT, IB	15 min
(b)	• Detailed Analysis of Mechanical Linkages • Hydraulic Boost Mechanisms in Power Assist • Electronic Signal Processing in Fly-by-Wire • Advantages and Disadvantages of Each Type • Integration of Control Surfaces	L & Ds	PPT, PDF	20 min
(c)	• Evolution from Mechanical to Digital Systems • Future Trends in Control System Technology • Maintenance Implications of Different Systems • Safety Protocols for Control Systems • Operational Limitations of Each Type	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a power-assisted and a fully powered control system.
(b) Describe how a fly-by-wire system replaces traditional mechanical linkages.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from mechanical to fly-by-wire systems to be discussed in LP 22 regarding Teleflex and Bowden cables.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Mechanical remote controls system: Teleflex, Bowden and use of control tube and cables	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min